

ALGORITHM

1) Start

2) Prompt the user for expected vehicles' quantity and store it in a variable 'N'

3) While N is greater than 0:

- Ask for emergency status. If the vehicle is not for emergency, then ask for permit status:

- If permit is not available, display a rejection message with reason. Decrement N, increment vehicles processed and rejected, and "Continue".

- Else if permit status is an invalid input, inform the user to try again, and "Continue"

- Else if emergency status is an invalid input, inform the user to try again, and "Continue"

- Prompt the user to enter vehicle type and store it in a variable. Prompt the user to try again if the input was invalid.

- Prompt the user to enter user category and store it in a variable. Prompt the user to try again if the input was invalid.

- If user category is Faculty

- If vehicle type is bike or car

- * If zone A has at least 1 slot remaining

- o update occupied and remaining slots in zone A.

- o Display success message and remaining slots in zone A.

- Increment accepted

- If vehicle-type is car, increment cars, else increment bikes

- * Else display rejection message due to insufficient space, increment rejected

- Else

- * If zoneA has atleast 2 slots remaining

- Add 2 to occupancy of zoneA and update remaining slots of zoneA.

- Display success message and remaining capacity in zoneA.

- Increment accepted and increment vary.

- * Else

- Display rejection message due to insufficient space and increment rejected.

- Else if usercategory is student-

- If vehicle-type is bike or car

- * If zoneB has atleast 1 slot remaining

- update occupied and remaining slots in zoneB

- Display success message and remaining slots in zoneB

- Increment accepted

- If vehicle-type is car, increment cars, else increment bikes

- * Else display rejection message due to insufficient space, increment rejected

■ Else

- * If zone C has atleast two slots remaining
 - Add two to occupancy of zone C and update remaining slots of zone C
 - Display success message and remaining slots in zone C
 - Increment accepted and vars
- * Else display rejection message due to insufficient space and increment rejected
- Else if user category is guest
 - If vehicle type is car or bike
 - * If zone C has atleast 1 slot remaining
 - Update occupied and remaining slots of zone C
 - Display success message and remaining capacity in zone C
 - Increment accepted
 - If vehicle type is bike, increment bikes else increment cars
 - * Else display rejection message and increment rejected.

■ Else

- * if zone C has atleast 2 slots remaining
 - Add 2 to occupancy of zone C and update remaining slots of zone C
 - Display success message and remaining slots of zone C

Date: _____

◦ Increment accepted and vans.

* Else display rejection message due to insufficient space and increment rejected.

• Decrement N and increment vehicles processed.

4) Display total vehicles processed, total accepted and rejected vehicles, total bikes and cars and vans successfully parked, final occupancy across zone A, zone B and zone C, remaining slots in zone A, zone B and zone C.

5) Display zone with highest occupancy.

6) Check if the parking area is full, if yes then display a corresponding message.

7) Stop.